

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL ONE MODULE, SPRING SEMESTER 2015-2016

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

Time allowed 1.5 hours

Candidates must NOT start writing their answers until told to do so

Answer THREE out of FOUR questions

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: None

<Proposed Marking Scheme>

Annotations Used:

K - Knowledge;
C - Comprehension;
A - Application

1. Heuristic Search. (25 marks)

- (a) With reference to the components of the path evaluation function $f(n)$, describe the relationship of the A^* search with *uniform cost* search and *greedy best first* search. [6 marks]

A^* search uses an evaluation function $f(n)$ which combines $g(n)$, the cost to reach the node from the initial state, and $h(n)$, the cost to get from the current node to the goal, i.e. $f(n) = g(n) + h(n)$. [2 marks] - **K**

Uniform cost search only minimizes the path cost incurred so far, $g(n)$, i.e. it expands the node n with the lowest path cost, i.e. $f(n) = g(n)$. [2 marks] - **K**

Greedy search expands that node that is closest to the goal, on the ground that this is likely to lead to a solution quickly, i.e. $f(n) = h(n)$. [2 marks] - **K**

- (b) Both “effective branching factor” and “informedness” are terms commonly used to compare qualities of heuristic functions. Explain how these two terms are related. [6 marks]

If a heuristic h_1 is better than another heuristic h_2 , h_1 will be greater than h_2 , subjected to the condition that they are both permissible. [3 marks] - **C**

In this case h_1 is said to be better informed than h_2 and using h_1 will result in a smaller effective branching factor, meaning that less number of nodes will be expanded as compared to h_2 in search for a solution. [3 marks] - **C**

- (c) Figure 1 shows a state space diagram (not to scale) for a path finding problem. Demonstrate how A^* search can be used to find the *optimal path* from A to G, assuming that any successor that is the same as the current node's parent are not expanded.

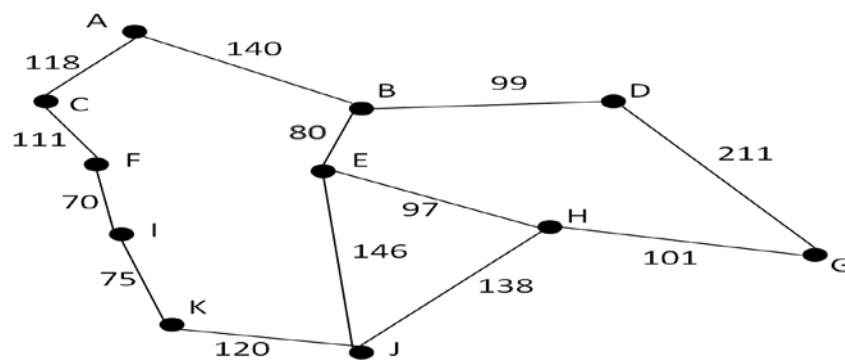


Figure 1.

The straight line distances from each node to G are given below:

A	366	G	0
B	253	H	100
C	329	I	241
D	176	J	160
E	193	K	242
F	244		

Your solution should include the following:

- (i) The status of the *fringe* (queue) at each step.

[4 marks]

[A366]

[B393, C447]

[E413, D415, C447]

[D415, H417, C447, J526]

[H417, C447, G450, J526]

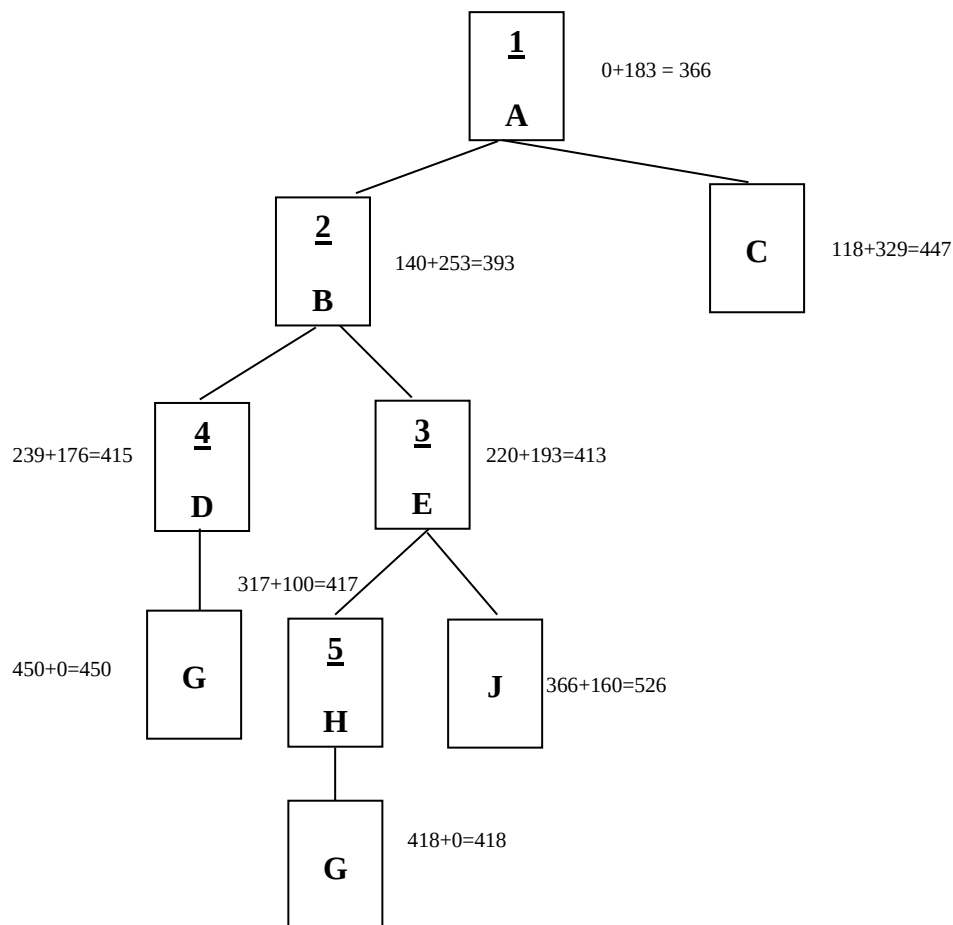
[G418, C447, G450, J526]

G is identified goal.

[4 marks] - A

- (ii) The *search tree* showing the order nodes are expanded, with their associated evaluation costs.

[4 marks]



[4 marks] - A

- (iii) The *shortest* route found and its path cost.

[2 marks]

The shortest route is A-B-E-H-G, path cost is 418. [3 marks] - A

- (iv) Explain how the optimal path from A to G is *constructed* by the search algorithm.

[3 marks]

Each node includes the information of how to link back to the parent node. [3 marks]

- C

2. Game Playing. (25 marks)

(a) Answer the following in the context of adversarial search:

- (i) Differentiate between *perfect information* and *imperfect information* games and name one example for each category. [4 marks]

Perfect information game means that the game is fully observable and both players have full access to all the states of the game. Acceptable examples are chess and checkers. [2 marks] - K

Imperfect information game is one that some information is hidden and only visible to one player. Acceptable examples are bridge and poker. [2 marks] - K

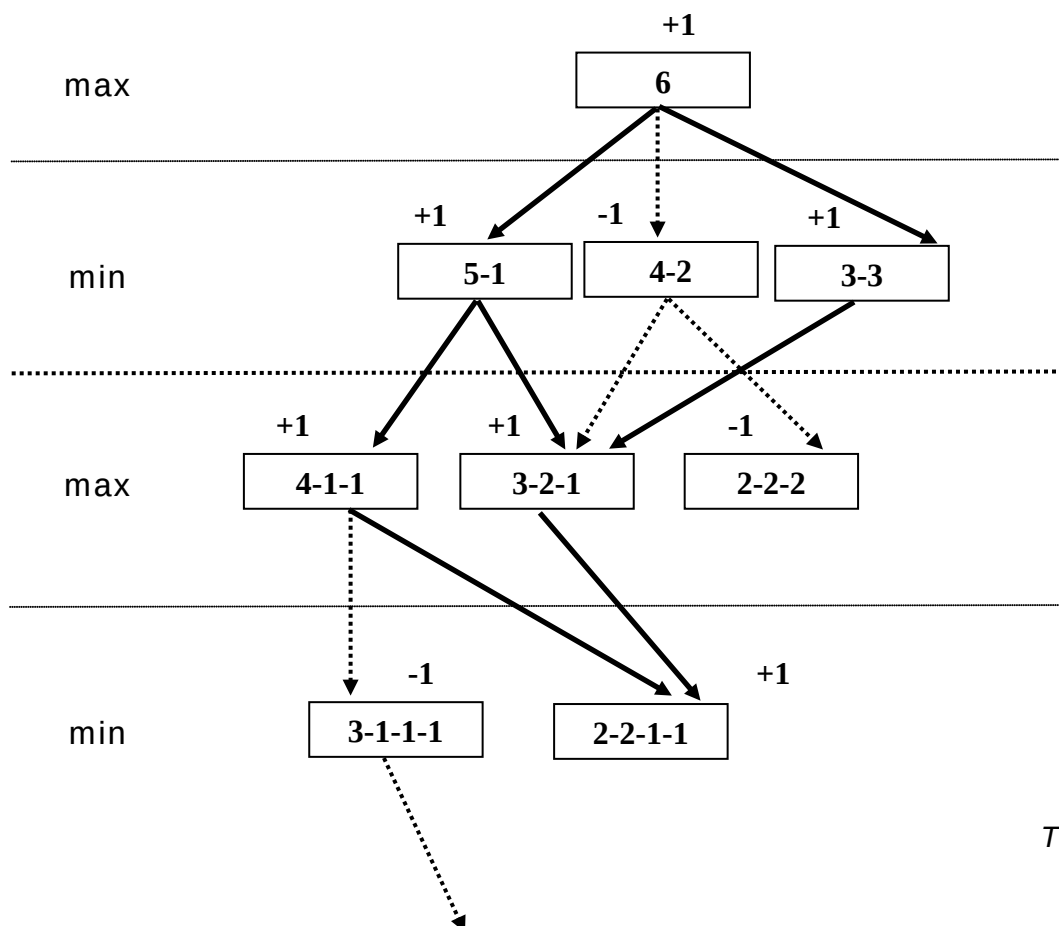
- (ii) What does it mean to say that a game has been solved? [3 marks]

When we say that a game has been solved, it means that the entire game tree can be built and outcome of the game can be determined even before the game is started. [3 marks] - C

- (b) Consider a game, "Xam6", that is a variation of the two player game Nim. Xam6 has the following rules:

The game starts with a single stack of 6 tokens. A valid move involves selecting one stack of three or more tokens and divides them into two non-empty stacks (unlike Nim the resulting stacks are allowed to be of equal size, except that they cannot both be size one). Two players alternate their moves and the first player unable to divide further the existing stack(s) loses the game.

- (i) Draw the complete *game tree* for Xam6. [8 marks]



	-1
max	2-1-1-1-1

[8 marks] - **A**

- (ii) Two players, min and max, play Xam6 with max making the first move. If a terminal state in the game tree developed above is a win for min, a utility function value of -1 is assigned to that state while a utility function value of +1 is assigned to a terminal state if it is a win for max. Apply the *minimax* algorithm to assign utility functions to all states in the game tree. [4 marks]

*The utility values are shown in the game tree above. Values should initially assigned to the terminal states and then propagated back to the root, depending on whether it is min or max's turn to move, as min will minimise the utility function and max will maximise. Player's turn should be marked correctly. [2 marks for correct marking of min and max turn, 2 marks for correct assignment of utility values] - **A***

- (iii) If both min and max play a perfect game, who will win? *Explain* your answer and *show* one possible path that can be taken by the player that wins. [3 marks]

*As the value at the root node is +1, max is guaranteed to win if it plays a perfect game. This is because min will never be able to follow a path that leads to a utility function of -1. [1 mark for explanation, 2 marks for showing a possible path] - **A***

- (c) What is the advantage of adding *alpha-beta pruning* to the minimax algorithm? [3 marks]

*It speeds up minimax algorithm by reducing the number of nodes that need to be examined, hence allowing n-ply look ahead to be made to a deeper depth. [3 marks] - **C***

3. Machine Learning. (25 marks)

- (a) In the context of machine learning, explain the difference between the following terms.

- (i) *Supervised and unsupervised learning* [4 marks]

*Supervised learning networks learn by being presented with some inputs along with their answers, which is called the training set. [2 marks] - **K***

*Unsupervised learning networks are never given training set and are basically left on their own to classify its inputs based on similarity metrics. [2 marks] - **K***

- (ii) Describe the difference between *k-fold* and *leave-one-out* cross validation methods for estimating accuracy of learning models. [6 marks]

k-fold cross validation method randomly partitions training data into k disjoint subsets, each roughly of size n/k where n is the total number of records. For example, if $k=10$, then 90% of the data is used for training, 10% of the data is used for testing and error measurement. This process is iterated 10 times, each time using a different 10% subset for testing. The final error is 1/10 of the total error found for the 10 iterations.

The leave-one-out cross validation method take k -fold to the extreme by setting $k=n$, and use $n-1$ data for training and 1 for testing. [6 marks] - C

- (b) A training set of 5 points are labelled with + and - using k -Nearest Neighbours (kNN) algorithm:

(1.5, +), (3.2, +), (5.4, -), (6.2, -), (8.5, -).

- (i) What is the predicted class for a test sample at point 4.0 using 1-NN based on Euclidean distance? [2 marks]

Closest neighbour is 3.2 so predicted class is +. [2 marks] - A

- (ii) What will be the predicted class for point 4.0 using 3-NN? [2 marks]

Closest neighbours are 3.2, 5.4 and 6.2. Predicted class is - since majority class is -. [2 marks] - A

- (c) Neurons in an artificial neural network learn by repeated adjustment of the weights associated with the connecting links.

- (i) State the *delta rule* for perceptron learning and describe the meaning of each term in the formula. [5 marks]

$$W_{j+1} = W_j + L_r * I_j * Err \text{ [2 marks] - K}$$

Err = T (training value) – O (actual output value);

I_j = presented input;

L_r = learning rate, rate of convergence;

W_j = current weight value;

W_{j+1} = adjusted new weight value. [3 marks] - K

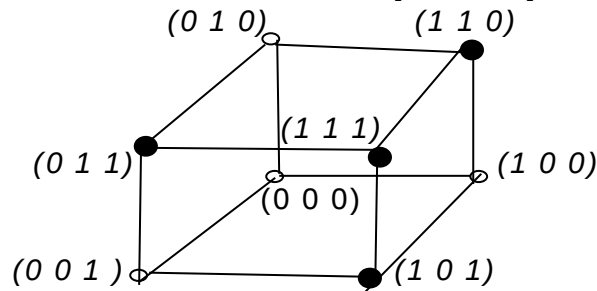
- (ii) A “Majority” function with three binary input values, gives an output of 1 if there are more 1's than 0's in the inputs, and 0 otherwise. Derive the truth table and state whether it is possible for a perceptron to learn this function or not. *Justify* your answer. [6 marks]

The Majority function can be represented by the following:

X	0	0	0	0	1	1	1	1
Y	0	0	1	1	0	0	1	1
Z	0	1	0	1	0	1	0	1
O	0	0	0	1	0	1	1	1

Correct truth table [3 marks] - A

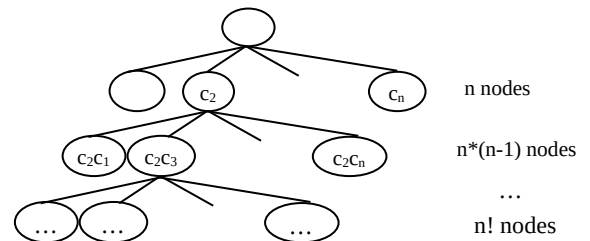
The "Majority" function can be represented by a single layer perceptron since it is linearly separable as illustrated below. [3 marks] – **A**



4. Knowledge Representation and Probabilistic Reasoning. (25 marks)

- (a) Use the Travelling Salesman Problem (TSP) as an example, explain what is meant by the term "combinatorial explosion". [3 marks]

This is known as combinatorial explosion where the number of solutions rises exponentially and the problem becomes intractable although theoretically it can be solved. [3 marks] – **K**



- (b) The process of building a knowledge based system (KBS) is known as knowledge engineering.

- (i) List and briefly describe its three main phases. [6 marks]

Knowledge acquisition: the process of getting the knowledge from experts (by interviewing and/or observing human experts, reading specific books, etc). [2 marks] – **K**

Knowledge representation: selecting the most appropriate structures to represent the knowledge. [2 marks] – **K**

Knowledge validation: testing that the knowledge coded is correct and complete. [2 marks] – **K**

- (ii) Translate the following first order predicate logic (FOPL) statement into English:

$$\forall (x, y, l) : \text{SpeaksLanguage}(x, l) \wedge \text{SpeaksLanguage}(y, l) \rightarrow \text{Understands}(x, y) \wedge \text{Understands}(y, x) \quad [4 \text{ marks}]$$

For any two people, if they speak the same language, they can understand each other well. [4 marks] – **A**

- (c) Student A tells his professor that he forgot to submit his assignment. From past experience, the professor knows that students who finish their assignment on time forget to turn it in 1 in 100 times. He also knows that half the students who have not completed their assignments will tell him they forgot to submit. He thinks that 90% of the students in his class finished their assignments on time.

Using Bayes' theorem defined below:

$$P(H_i|E) = \frac{P(E|H_i) \times P(H_i)}{\sum_{n=1}^N P(E|H_n) \times P(H_n)}$$

- (i) What is the *prior* probability that a student forget to submit his/her assignment? [6 marks]

Let Forget be the variable indicating that the student forgot to submit assignment; and Finish be the variable indicating that the student's assignment was finished on time.

$$P(\text{Forget}) = P(\text{Forget} | \text{Finish}) * P(\text{Finish}) + P(\text{Forget} | \neg \text{Finish}) * P(\neg \text{Finish})$$

$$P(\text{Forget}) = 0.01 * 0.9 + 0.5 * 0.1 = 0.059 \text{ [6 marks] - A}$$

- (ii) What is the probability that student A is telling the *truth*, i.e. he/she finished the assignment but forgot to submit? [6 marks]

$$P(\text{Finish} | \text{Forget}) = P(\text{Forget} | \text{Finish}) * P(\text{Finish}) / P(\text{Forget})$$

$$P(\text{Finish} | \text{Forget}) = 0.01 * 0.9 / 0.059 = 0.1525 \text{ [6 marks] - A}$$